

# Web scrapping for books

## Step 1: Import Libraries and Fetch the Page

```
In [13]: import requests
         from bs4 import BeautifulSoup
         import pandas as pd
```

## Step 2: Send an HTTP Request and Check the Response

```
In [14]: url = 'https://books.toscrape.com/'
         response = requests.get(url)

         # Check if request was successful (200 means success)
         print("Status Code:", response.status_code)
```

Status Code: 200

## Step 3: Parse the HTML with BeautifulSoup

Convert the raw HTML text into a structured BeautifulSoup object so we can navigate tags.

```
In [15]: soup = BeautifulSoup(response.text, 'html.parser')

         # Preview the title of the webpage
         print("Page Title:", soup.title.text.strip())
```

Page Title: All products | Books to Scrape - Sandbox

## Step 4: Inspect and Extract Data for a Single Book

On the website, every book is enclosed in an

tag with the class product\_pod. Let's isolate the first book to test our extraction logic.

```
In [16]: # From HTML: <article class="product_pod">
         book = soup.find('article', class_='product_pod')

         # From HTML: <a href="..." title="A Light in the Attic">A Light in the Attic</a>
         title = book.h3.a['title']

         # From HTML: <p class="price_color">£51.77</p>
         price = book.find('p', class_='price_color').text

         # From HTML: <p class="star-rating Three"></p>
         rating = book.find('p', class_='star-rating')['class'][1] # extracts "Three" aft

         print(f"Title: {title} | Price: {price} | Rating: {rating}")
```

Title: A Light in the Attic | Price: £51.77 | Rating: Three

## Step 5: Loop Through All Books on the Page

```
In [17]: books_data = []

# From HTML: <article class="product_pod"> (Repeated for every book listed)
books = soup.find_all('article', class_='product_pod')

for b in books:
    # From HTML: <a title="Book Title">
    title = b.h3.a['title']

    # From HTML: <p class="price_color">£XX.XX</p>
    price = b.find('p', class_='price_color').text

    # From HTML: <p class="star-rating RatingName">
    rating = b.find('p', class_='star-rating')['class'][1]

    books_data.append({
        'Title': title,
        'Price': price,
        'Rating': rating
    })
```

## Step 6: Convert and Display in Pandas DataFrame

```
In [18]: df = pd.DataFrame(books_data)

# Display the first 5 rows in Jupyter Lab
df
```

Out[18]:

|    | Title                                             | Price   | Rating |
|----|---------------------------------------------------|---------|--------|
| 0  | A Light in the Attic                              | Â£51.77 | Three  |
| 1  | Tipping the Velvet                                | Â£53.74 | One    |
| 2  | Soumission                                        | Â£50.10 | One    |
| 3  | Sharp Objects                                     | Â£47.82 | Four   |
| 4  | Sapiens: A Brief History of Humankind             | Â£54.23 | Five   |
| 5  | The Requiem Red                                   | Â£22.65 | One    |
| 6  | The Dirty Little Secrets of Getting Your Dream... | Â£33.34 | Four   |
| 7  | The Coming Woman: A Novel Based on the Life of... | Â£17.93 | Three  |
| 8  | The Boys in the Boat: Nine Americans and Their... | Â£22.60 | Four   |
| 9  | The Black Maria                                   | Â£52.15 | One    |
| 10 | Starving Hearts (Triangular Trade Trilogy, #1)    | Â£13.99 | Two    |
| 11 | Shakespeare's Sonnets                             | Â£20.66 | Four   |
| 12 | Set Me Free                                       | Â£17.46 | Five   |
| 13 | Scott Pilgrim's Precious Little Life (Scott Pi... | Â£52.29 | Five   |
| 14 | Rip it Up and Start Again                         | Â£35.02 | Five   |
| 15 | Our Band Could Be Your Life: Scenes from the A... | Â£57.25 | Three  |
| 16 | Olio                                              | Â£23.88 | One    |
| 17 | Mesaerion: The Best Science Fiction Stories 18... | Â£37.59 | One    |
| 18 | Libertarianism for Beginners                      | Â£51.33 | Two    |
| 19 | It's Only the Himalayas                           | Â£45.17 | Two    |

```
In [21]: # Use ExcelWriter with mode='a' to open the existing file
# if_sheet_exists='replace' will update/overwrite just the "Books" sheet inside the
excel_filename = "Scrapped.xlsx"
with pd.ExcelWriter(excel_filename, engine='openpyxl', mode='a', if_sheet_exists='r
    df.to_excel(writer, index=False, sheet_name="Books")

print(f"Data successfully updated inside existing file {excel_filename}!")
```

Data successfully updated inside existing file Scrapped.xlsx!